

ML BASED STRABISMUS RISK ASSESSMENT SYSTEM

A PROJECT REPORT

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We hereby declare that the work entitled “**ML BASED STRABISMUS RISK ASSESSMENT SYSTEM**” is submitted in partial fulfillment of the requirement for the award of the degree in B.Tech, in University College of engineering, BIT Campus, Anna University, Tiruchirappalli. It is the record of my own work carried out during the academic year 2025 – 2026 under the supervision and guidance of **Mr.M.KRISHNA KUMAR**, Guest Faculty, Department of Information Technology, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli. The extent and source of information are derived from the existing literature and have been indicated through the dissertation at the appropriate places. The matter embodied in this work is original and has not been submitted for the award of any other degree, either in this or any other university.

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ABSTRACT

Strabismus is a common vision disorder characterized by the misalignment of the eyes, which can lead to impaired vision and long-term complications if not detected early. Traditional diagnostic methods rely heavily on clinical examination by specialists, making early detection difficult in remote or resource-limited settings. To address this challenge, this project proposes a Machine Learning (ML)-Based Strabismus Risk Assessment System that integrates image processing and clinical data analysis to provide an automated, efficient, and user-friendly solution.

The proposed system utilizes frontal eye images along with clinical inputs such as age, onset type, presence of double vision, headache, diabetes, and thyroid conditions. The image data is processed using a Convolutional Neural Network (CNN) model to detect strabismus features, while the clinical data is analyzed using a Logistic Regression model to estimate risk probability. A fusion mechanism combines both outputs to generate a final risk classification, improving prediction accuracy and reliability.

The system is implemented as a web-based application where users can upload or capture eye images and input clinical details. The application performs real-time analysis and provides outputs including detection results, risk level (low, moderate, or high), reasons based on input data, and recommendations for further medical consultation. The model achieved an accuracy of approximately 96%, demonstrating strong performance in classifying normal and strabismus cases.

This approach enhances early detection, reduces dependency on manual diagnosis, and makes screening accessible to a wider population. The proposed system can serve as a supportive tool for healthcare professionals and can be extended for real-time clinical applications. Overall, the project contributes to improving eye health management through the integration of artificial intelligence and medical data analysis.

சுருக்கம்

Strabismus என்பது கண்கள் ஒரே திசையில் நிலையாக இல்லாமல் தவறாக சாய்வதைக் குறிக்கும் ஒரு பொதுவான கண் குறைபாடாகும். இது ஆரம்ப நிலையில் கண்டறியப்படாமல் விட்டால் பார்வை குறைபாடு மற்றும் நீண்டகால பாதிப்புகளை ஏற்படுத்தும். வழக்கமாக, இந்த நோயை கண்டறிவது நிபுணர்கள் மேற்கொள்ளும் மருத்துவ பரிசோதனைகளின் மூலம் மட்டுமே நடைபெறுகிறது. இதனால், தொலைதூர பகுதிகளில் அல்லது மருத்துவ வசதி குறைவான இடங்களில் ஆரம்ப கட்ட கண்டறிதல் சிரமமாகிறது. இந்த சிக்கலை தீர்க்க, இத்திட்டம் Machine Learning அடிப்படையிலான Strabismus Risk Assessment System ஒன்றை முன்வைக்கிறது.

இந்த அமைப்பு கண் முன்னோக்கிய படங்கள் மற்றும் மருத்துவ தகவல்கள் (வயது, நோய் தொடங்கிய வகை, இரட்டை பார்வை, தலைவலி, நீரிழிவு, தைராய்டு பிரச்சினைகள் போன்றவை) ஆகியவற்றை பயன்படுத்துகிறது. கண் படங்களை Convolutional Neural Network (CNN) மூலம் பகுப்பாய்வு செய்து strabismus இருப்பதை கண்டறிகிறது. மருத்துவ தகவல்கள் Logistic Regression முறையின் மூலம் அபாய நிலையை கணிக்கப் பயன்படுத்தப்படுகின்றன. இந்த இரண்டு முடிவுகளையும் ஒன்றிணைக்கும் fusion முறையின் மூலம் இறுதி முடிவு உருவாக்கப்படுகிறது.

இந்த அமைப்பு ஒரு web application ஆக உருவாக்கப்பட்டுள்ளது. பயனர்கள் தங்கள் கண் படங்களை பதிவேற்றவோ அல்லது கேமரா மூலம் பிடிக்கவோ முடியும். அதன்பின் மருத்துவ தகவல்களை வழங்கி உடனடியாக முடிவுகளைப் பெறலாம். வெளியீடாக strabismus கண்டறிதல், அபாய நிலை (குறைவு, மிதமானது, அதிகம்), காரணங்கள் மற்றும் மருத்துவ ஆலோசனைகள் வழங்கப்படுகின்றன. இந்த முறை சுமார் 96% துல்லியத்துடன் செயல்படுகிறது.

இந்த அமைப்பு ஆரம்ப கட்ட நோய் கண்டறிதலை எளிதாக்குகிறது, கையால் செய்யும் பரிசோதனைகளின் தேவையை குறைக்கிறது மற்றும் அதிகமான மக்களுக்கு அணுகக்கூடியதாக உள்ளது. இது மருத்துவ நிபுணர்களுக்கு உதவும் கருவியாக செயல்படுவதுடன், எதிர்காலத்தில் நேரடி மருத்துவ பயன்பாடுகளுக்கு விரிவுபடுத்த முடியும். மொத்தத்தில், இந்த திட்டம் செயற்கை நுண்ணறிவு மற்றும் மருத்துவ தரவு பகுப்பாய்வை இணைத்து கண் ஆரோக்கிய மேலாண்மையை மேம்படுத்துகிறது.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

SYMBOL / TERM	DESCRIPTION
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
LR	Logistic Regression
AI	Artificial Intelligence
CV	Computer Vision
UI	User Interface
UX	User Experience
API	Application Programming Interface
DB	Database
SQL	Structured Query Language
SQLite	Lightweight Relational Database
Flask	Python Web Framework
NumPy	Numerical Python Library
Pandas	Data Analysis Library
Matplotlib	Data Visualization Library
OpenCV	Open Source Computer Vision Library
TensorFlow	Deep Learning Framework
Keras	High-Level Neural Network API
MobileNetV2	Lightweight CNN Architecture
RGB	Red Green Blue (Image Format)
JPEG	Joint Photographic Experts Group
PNG	Portable Network Graphics
ROI	Region of Interest
ReLU	Rectified Linear Unit
Softmax	Activation Function for Classification
Epoch	One Complete Training Cycle
Batch Size	Number of Samples per Training Step
Accuracy	Measure of Correct Predictions
Precision	Measure of Positive Prediction Accuracy
Recall	Measure of Model Sensitivity
F1-Score	Harmonic Mean of Precision and Recall
Confusion	Performance Evaluation Table

Matrix	
Dataset	Collection of Training Data
Training Data	Data Used to Train Model
Validation Data	Data Used to Tune Model
Testing Data	Data Used to Evaluate Model
Strabismus	Eye Misalignment Condition
Diplopia	Double Vision
Risk Score	Probability of Disease Occurrence
Normalization	Scaling Input Data
Encoding	Converting Data into Machine Format
Preprocessing	Data Preparation Step
Inference	Model Prediction Phase

CHAPTER 1

INTRODUCTION

Strabismus is a common ocular disorder characterized by the misalignment of the eyes, where both eyes do not focus on the same direction simultaneously. If not detected and treated at an early stage, it can lead to serious complications such as double vision, poor depth perception, and amblyopia (lazy eye). Early diagnosis of strabismus is therefore essential to prevent long-term visual impairment and improve the quality of life of affected individuals.

In recent years, the advancement of Artificial Intelligence (AI) and Machine Learning (ML) has significantly transformed the field of medical diagnosis. Deep learning techniques, particularly Convolutional Neural Networks (CNN), have shown promising results in analyzing medical images with high accuracy. However, most existing systems focus only on image-based detection and do not consider clinical symptoms and patient-related factors, which play a crucial role in real-world diagnosis.

This project aims to develop a Machine Learning-based strabismus risk assessment system that integrates both image-based analysis and clinical questionnaire data. The proposed system utilizes a MobileNetV2 model for eye image classification and Logistic Regression for analyzing clinical risk factors. By combining these two approaches through a hybrid scoring mechanism, the system provides a more reliable and practical solution for early screening and public awareness.

The system is designed to be user-friendly and accessible through a web-based interface, allowing users to upload eye images and input clinical details easily. The final output includes strabismus detection, risk level classification, and recommendation, thereby assisting users in taking timely medical action. This approach reduces dependency on manual diagnosis and contributes to the development of cost-effective and scalable healthcare solutions.

1.1 BACKGROUND

Strabismus is a visual disorder characterized by the misalignment of the eyes, where one eye may turn inward, outward, upward, or downward while the other eye focuses normally. This condition affects both children and adults and can significantly impact visual coordination. If left untreated, strabismus may lead to serious complications such as amblyopia (lazy eye), double vision (diplopia), reduced depth perception, and impaired binocular vision.

Traditional methods for diagnosing strabismus include clinical tests such as the Hirschberg test, cover-uncover test, and prism cover test. These methods rely heavily on the expertise and experience of ophthalmologists. While these techniques are effective, they are time-consuming,

subjective, and require specialized clinical environments. Additionally, regular screening is not always accessible, especially in rural and remote areas where medical facilities are limited.

Early detection and diagnosis of strabismus are critical in preventing permanent visual impairment and ensuring effective treatment. With the increasing prevalence of eye disorders and the growing need for accessible healthcare solutions, there is a strong demand for automated diagnostic systems.

Advancements in Artificial Intelligence (AI) and Machine Learning (ML), particularly in deep learning, have opened new possibilities in medical image analysis. Convolutional Neural Networks (CNN) have demonstrated high accuracy in detecting patterns from visual data, making them suitable for eye disease detection. However, real-world diagnosis not only depends on visual data but also on clinical symptoms and patient history.

Therefore, integrating image-based analysis with clinical data can significantly enhance the accuracy and reliability of diagnostic systems. This motivates the development of a hybrid system that combines both approaches to support early screening and improve healthcare accessibility.

1.2 PROBLEM STATEMENT

Strabismus is often not diagnosed at an early stage due to lack of awareness, absence of regular screening, and dependency on manual clinical examination. Many individuals ignore early symptoms such as double vision, eye strain, or sudden onset of eye deviation, which can lead to delayed treatment and long-term visual complications. Early-stage detection is crucial, but it is not easily accessible to the general public without visiting specialized healthcare facilities.

Although recent developments in Artificial Intelligence have introduced automated systems for strabismus detection, most existing solutions focus only on image-based analysis using deep learning models. These systems do not incorporate clinical or patient-specific information such as symptoms, medical history, or risk factors, which are essential for accurate diagnosis.

Furthermore, many existing approaches require high-quality medical imaging equipment or controlled environments, making them impractical for real-time or public usage. Some systems are computationally complex and not optimized for lightweight or web-based applications. In addition, the lack of integration between multiple data sources limits the overall reliability of these systems.

Hence, there is a need for a cost-effective, user-friendly, and hybrid machine learning system that combines both eye image analysis and clinical questionnaire data. Such a system can provide early risk assessment, improve accessibility, and support public awareness by enabling users to perform preliminary screening without the need for specialized equipment.

1.3 OBJECTIVES

The main objective of this project is to develop an intelligent system for early detection and risk assessment of strabismus using machine learning techniques.

Specific Objectives:

- To develop an image-based detection model using MobileNetV2 for identifying strabismus.
- To design a questionnaire-based system using Logistic Regression for analyzing clinical risk factors.
- To integrate both image and clinical predictions using a hybrid scoring mechanism.
- To classify the risk level into Low, Moderate, and High categories.
- To build a user-friendly web-based interface for public accessibility and awareness.
- To improve early screening and reduce dependency on manual diagnosis.

CHAPTER 2

2.1 INTRODUCTION TO LITERATURE REVIEW

The literature review provides an overview of existing research and methodologies related to strabismus detection and ocular disease analysis using Artificial Intelligence and Machine Learning techniques. Recent advancements in deep learning, particularly Convolutional Neural Networks (CNN), have enabled automated analysis of eye images with high accuracy.

Several studies have focused on image-based detection of strabismus using facial or retinal images. Pretrained models such as MobileNet, ResNet, and VGG have been widely used due to their ability to extract meaningful features from visual data. However, most existing systems depend only on image-based inputs and do not consider clinical or patient-specific factors.

Few research works have explored the integration of clinical assessment methods such as cover test, eye alignment analysis, and symptom-based evaluation. However, these systems are either manual or lack automation and scalability.

Therefore, a hybrid approach combining deep learning-based image analysis with questionnaire-based clinical assessment can provide a more reliable, accessible, and practical solution for early strabismus detection and risk assessment.

2.2 REVIEW OF EXISTING SYSTEMS AND PAPERS

Paper 1: Tianlun Wang et al. (2024, IEEE)

Title: Mobile Strabismus Automatic Detection Method Combining Alternating Cover Test and Deep Learning

Summary:

This paper presents a mobile-based strabismus detection system that integrates the alternating cover test with deep learning techniques. The system captures eye movement patterns using a smartphone camera and analyzes them using a Convolutional Neural Network (CNN). It aims to provide a portable and accessible solution for early diagnosis. The combination of clinical testing and AI improves detection accuracy. The system is designed to support remote screening and preliminary diagnosis.

Limitations:

The system requires a controlled environment for accurate results. It depends heavily on proper user positioning and camera alignment. The dataset used is relatively small, which affects model

generalization. Performance may degrade in real-world conditions. It lacks robustness against lighting variations and noise.

Paper 2: Mahsa Yarkheir et al. (2025, Scientific Reports)

Title: Automated Strabismus Detection and Classification using Deep Learning Analysis of Facial Images

Summary:

This study proposes a deep learning-based approach for detecting strabismus using facial images. A CNN model is trained to extract features from the eye region and classify alignment abnormalities. The system demonstrates high accuracy in controlled datasets. It focuses on non-invasive image-based diagnosis without requiring specialized equipment. The method shows promise for automated screening applications.

Limitations:

The model requires large amounts of labeled data for effective training. Performance is highly dependent on image quality. It struggles under poor lighting and noisy conditions. Real-world generalization is limited. The system may not perform consistently across diverse populations.

Paper 3: Madior Gueye (2026, Springer)

Title: Comparative Study of Deep Learning Models for Strabismus Diagnosis using Facial Images

Summary:

This paper compares different deep learning models such as CNN, MobileNet, and ResNet for detecting strabismus. It evaluates performance metrics including accuracy, precision, and recall across models. The study highlights strengths and weaknesses of each architecture. It provides insights into selecting suitable models for medical image classification. The research emphasizes the importance of model optimization.

Limitations:

The results vary significantly depending on dataset quality. There is no standardized dataset for consistent comparison. Some models require high computational resources. The study lacks real-time implementation analysis. Generalization across different datasets remains a challenge.

Paper 4: Alaa Daher et al. (2023, IEEE)

Title: Artificial Intelligence-Powered System for Detecting, Diagnosing, and Rehabilitating Strabismus Disorder

Summary:

This paper introduces an AI-powered system that covers detection, diagnosis, and rehabilitation of strabismus. It integrates multiple modules for analyzing eye alignment and suggesting treatment options. The system aims to provide a comprehensive solution for both diagnosis and recovery. It combines clinical methods with intelligent algorithms. The approach improves decision-making support for healthcare professionals.

Limitations:

The system is complex and involves multiple components. It requires significant computational resources. Implementation is not suitable for lightweight or web-based platforms. Integration and maintenance are challenging. Real-time usability is limited.

Paper 5: Govardhana Rekam (2026, Springer)

Title: AI-Enabled Automated Strabismus Detection and Corneal Deviation Analysis

Summary:

This work focuses on detecting strabismus by analyzing corneal deviation using AI and image processing techniques. The system identifies alignment differences in eye positions through image analysis. It enhances accuracy by combining feature extraction and classification methods. The approach provides a more detailed understanding of eye misalignment. It contributes to improving diagnostic precision.

Limitations:

The study lacks validation in real-world scenarios. It requires more diverse and large-scale datasets. Performance may vary across different imaging conditions. The system is sensitive to image quality. Further optimization is needed for practical deployment.

Paper 6:

Title: Orthoptic Assessment of a Patient with Strabismus

Summary:

This paper discusses traditional orthoptic assessment methods used for diagnosing strabismus. It focuses on clinical techniques such as the cover test and eye alignment evaluation. The study highlights the importance of expert observation in diagnosis. It provides insights into manual diagnostic procedures. The approach is widely used in clinical practice.

Limitations:

The method is completely manual and lacks automation. It depends heavily on clinical expertise.

The process is time-consuming and not scalable. It is not suitable for remote or early screening. No integration with modern AI techniques is provided.

2.3 COMPARATIVE ANALYSIS

Table 2.3.1

Paper / Study	Method Used	Tools / Techniques	Accuracy / Performance	Key Observation
Tianlun Wang et al. (2024)	Deep Learning + Alternating Cover Test	CNN, Mobile-based vision system	High (~90–95%)	Combines clinical test with AI but needs controlled setup
Mahsa Yarkheir et al. (2025)	Deep Learning (CNN)	CNN, Image Processing	High (~92–96%)	Good performance but depends on image quality
Springer Study (2026)	Comparative DL Models	CNN, MobileNet, ResNet	High (varies)	Performance varies based on model and dataset
Alaa Daher et al. (2023)	AI-based Integrated System	AI modules, Multi-stage processing	Moderate–High	Complex system, not suitable for lightweight use
Springer Study (2026)	Image Processing + AI	Corneal deviation analysis, ML	Moderate–High	Needs real-world validation and larger datasets
Springer Study (2024)	Traditional Clinical Methods	Cover test, Eye alignment analysis	Moderate (~80–85%)	No automation, depends on experts
Proposed System (2026)	CNN + Logistic Regression (Hybrid ML)	TensorFlow, OpenCV, Scikit-learn, Flask	Very High (~96%)	Combines image + clinical data for accurate prediction

The comparative analysis of existing systems highlights that traditional clinical methods and basic machine learning approaches provide only moderate accuracy and heavily depend on expert intervention. In contrast, deep learning-based methods such as Convolutional Neural Networks (CNN), MobileNet, and ResNet demonstrate significantly improved performance, achieving accuracy levels above 90% in most cases. Studies like Tianlun Wang et al. and Mahsa Yarkheir et al. show that integrating image-based analysis with deep learning enhances detection capability, although these systems often require controlled environments and high-quality datasets. Comparative studies further indicate that model performance varies depending on the dataset and architecture used, emphasizing the need for robust and generalized models.

Additionally, AI-integrated systems such as the one proposed by Alaa Daher et al. offer comprehensive solutions including detection and rehabilitation, but their complexity limits real-time and lightweight deployment. Image processing-based approaches focusing on corneal deviation provide useful insights but lack sufficient real-world validation. Traditional orthoptic methods, while clinically reliable, are time-consuming and lack automation. Based on these observations, the proposed system improves upon existing methods by combining CNN-based image analysis with Logistic Regression-based clinical data evaluation. This hybrid approach enhances prediction accuracy, reduces dependency on controlled environments, and enables real-time, user-friendly implementation through a web-based platform.

CHAPTER 3

3.1 Existing System

Existing systems for strabismus detection primarily rely on either traditional clinical examination methods or image-based machine learning techniques. Conventional clinical approaches such as the Hirschberg test, cover-uncover test, and prism cover test are widely used by ophthalmologists to diagnose eye misalignment. These methods involve manual observation and require expert knowledge to interpret the results accurately. In recent years, automated detection systems have been introduced using deep learning techniques, particularly Convolutional Neural Networks (CNN). These systems analyze facial or eye images to identify patterns of misalignment and classify the presence of strabismus. Some advanced approaches also utilize mobile-based image capture to enable real-time analysis and remote screening. However, most of these systems rely solely on image data and do not incorporate additional clinical or patient-related information, which limits their overall effectiveness.

Despite these advancements, existing systems face several significant limitations. The dependence on only image-based analysis reduces diagnostic reliability, as it ignores important clinical factors such as symptoms and medical history. Many systems require high-quality images captured under controlled lighting and positioning conditions, making them less practical for real-world use. Additionally, certain approaches depend on specialized medical equipment, increasing cost and limiting accessibility for general users. Deep learning models often involve high computational complexity, making them difficult to deploy in lightweight or web-based applications. Furthermore, traditional clinical methods are time-consuming and subjective, as they depend heavily on the expertise of medical professionals. These limitations highlight the need for a more efficient, integrated, and user-friendly system for accurate strabismus detection.

3.2 Proposed System

The proposed system is a Machine Learning-based Strabismus Risk Assessment System designed to provide early detection and accurate risk classification using a hybrid approach. The system integrates image-based analysis using a deep learning model (MobileNetV2) and clinical questionnaire analysis using Logistic Regression. Users can upload or capture an eye image and provide clinical details such as age, onset type, and symptoms including double vision, headache, diabetes, and thyroid conditions. The system processes both inputs independently, where the image is analyzed using the CNN model to detect eye misalignment, and the clinical data is evaluated to estimate risk probability. These predictions are then combined using a rule-based threshold mechanism to improve reliability. If the image detects strabismus and either the questionnaire risk is not low or the image confidence is very high, the system confirms the presence of strabismus. The final output includes the detection result, risk classification (low, moderate, or high), along

with reasons and recommendations. The entire system is implemented as a web-based application, enabling easy access for users and supporting early screening.

The proposed system offers several advantages over existing approaches. By combining both image-based and clinical data, it improves prediction accuracy and reduces the chances of misclassification. The system enables early risk assessment, which helps in timely medical intervention and better eye health management. Its user-friendly web interface makes it accessible to the general public without requiring specialized medical equipment. The use of MobileNetV2 ensures that the system remains lightweight and efficient, making it suitable for real-time applications. Additionally, the system reduces dependency on manual clinical diagnosis and minimizes subjectivity in decision-making. It is cost-effective, scalable, and capable of reaching a wider population. Furthermore, the system provides clear and interpretable outputs, including reasons and recommendations, which enhance user understanding and trust in the results.

3.3 SYSTEM ARCHITECTURE

The proposed ML-Based Strabismus Risk Assessment System is designed as a modular and distributed architecture that integrates both image-based deep learning and clinical data analysis within a web-based environment. The system follows a layered processing model in which each component performs a specific function, and together they provide an accurate and reliable risk assessment for strabismus detection. The overall data flow follows a structured pipeline: the user provides input in the form of a frontal face image and clinical questionnaire data, the web application collects and validates the inputs, preprocessing is performed on both image and clinical data, the processed inputs are passed to respective machine learning models, and the outputs are fused using rule-based logic to generate the final result. At each stage, the system ensures proper data transformation, validation, and prediction to maintain consistency and reliability in the assessment process.

The architecture is organized into multiple functional layers. The input layer handles user-provided data including image upload and questionnaire responses. The application layer (web application) manages data collection, input validation, and feature scaling to ensure that the data is in a suitable format for further processing. The data processing layer performs image preprocessing such as resizing and normalization, along with questionnaire data preparation and feature extraction. The model inference layer consists of two independent models: a Convolutional Neural Network (CNN) for image-based strabismus detection and a questionnaire classifier (Logistic Regression) for clinical risk prediction. The fusion layer combines outputs from both models using rule-based decision logic to determine the final diagnosis and risk level. The output layer presents the detection result, risk classification, and recommendations to the user, while also storing the assessment records in a local SQLite database for future reference.

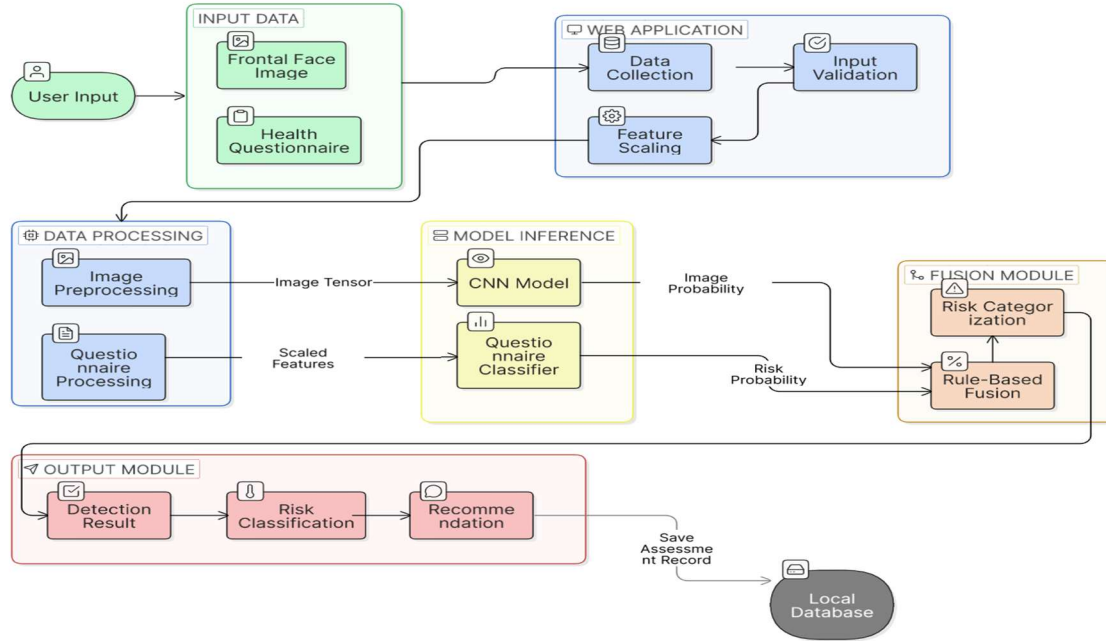


Fig 1 System Architecture

The component dependency structure is clearly defined: the web application interacts with preprocessing modules for data preparation, the CNN model depends on processed image tensors, and the questionnaire classifier depends on scaled clinical features. The fusion module integrates outputs from both models to generate a final decision. The output module depends on the fusion results to display meaningful insights to the user, and the database module stores all assessment records. This structured and layered architecture ensures scalability, modularity, and efficient execution of the system while maintaining accuracy and usability.

3.3.1 Data Flow Diagram

The data flow diagram illustrates how data moves through the proposed system from input to final output. Initially, the user provides input in two forms: an eye image and clinical questionnaire data through the web application interface. The uploaded image is validated and passed to the preprocessing module, where it is resized, normalized, and converted into a suitable format. The processed image is then fed into the MobileNetV2 deep learning model, which extracts features and predicts the presence of strabismus. At the same time, the questionnaire data is validated, encoded, and normalized to convert it into a structured numerical format for analysis.

The processed clinical data is then passed to the Logistic Regression model, which generates a risk probability based on user symptoms and medical conditions. The outputs from both models are combined using a rule-based decision mechanism to produce a final prediction. The system then

classifies the result into risk levels such as low, moderate, or high, along with detection status. The final output, including reasons and recommendations, is displayed to the user through the web interface. Additionally, all results are stored in a SQLite database for future reference and system monitoring, ensuring efficient data management and real-time performance.

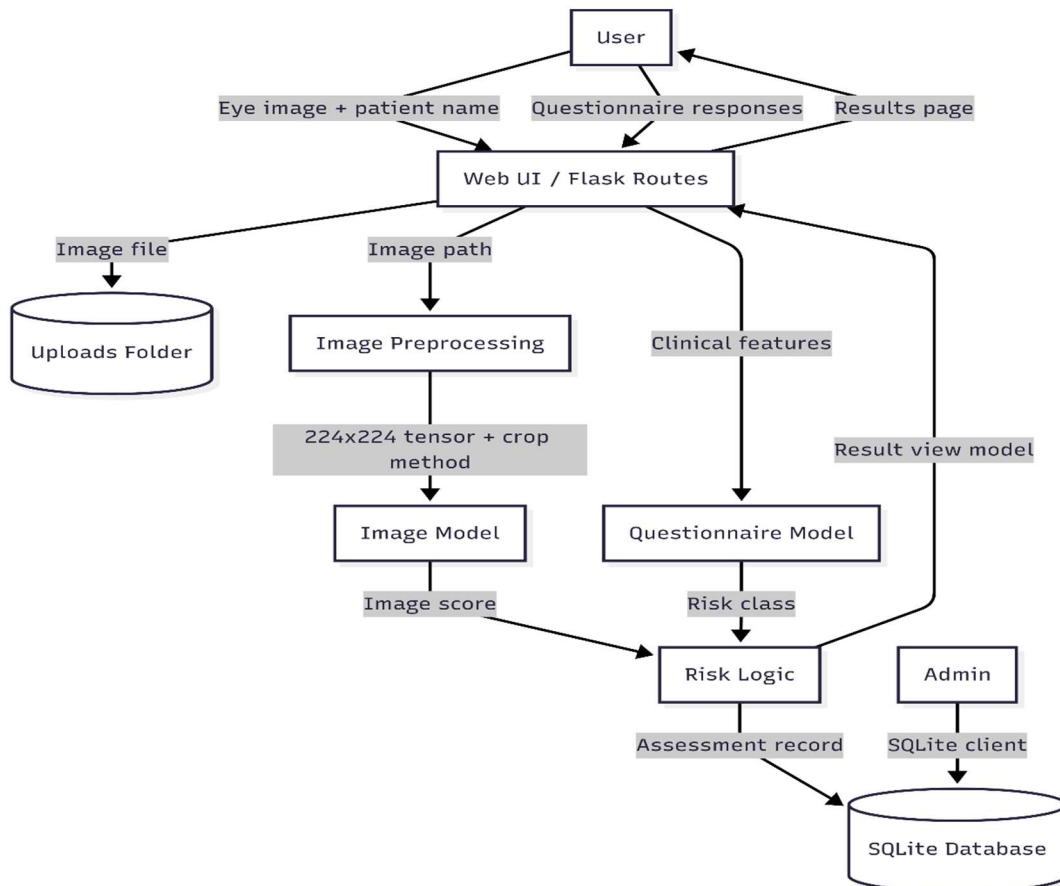


Fig 3 Dataflow Diagram

3.3.2 Use Case Diagram

The use case diagram represents the interaction between the main actors and the proposed system. The primary actor is the user, who interacts with the web application to perform the strabismus assessment. The user uploads or captures an eye image and provides clinical details through a questionnaire, including age and symptoms. Based on these inputs, the system processes the data by performing image analysis using the CNN model and clinical risk evaluation using the Logistic Regression model. After processing, the system generates and displays the final result, which includes the detection status, risk level, reasons, and recommendations.

In addition to the user, the system also involves an admin or clinician as a secondary actor. The system stores all assessment results in a SQLite database for future reference and analysis. The admin or researcher can access this stored data directly to review past assessments, monitor system performance, and support further improvements. This interaction ensures efficient data management while enabling both user-level access for prediction and admin-level access for analysis and evaluation.

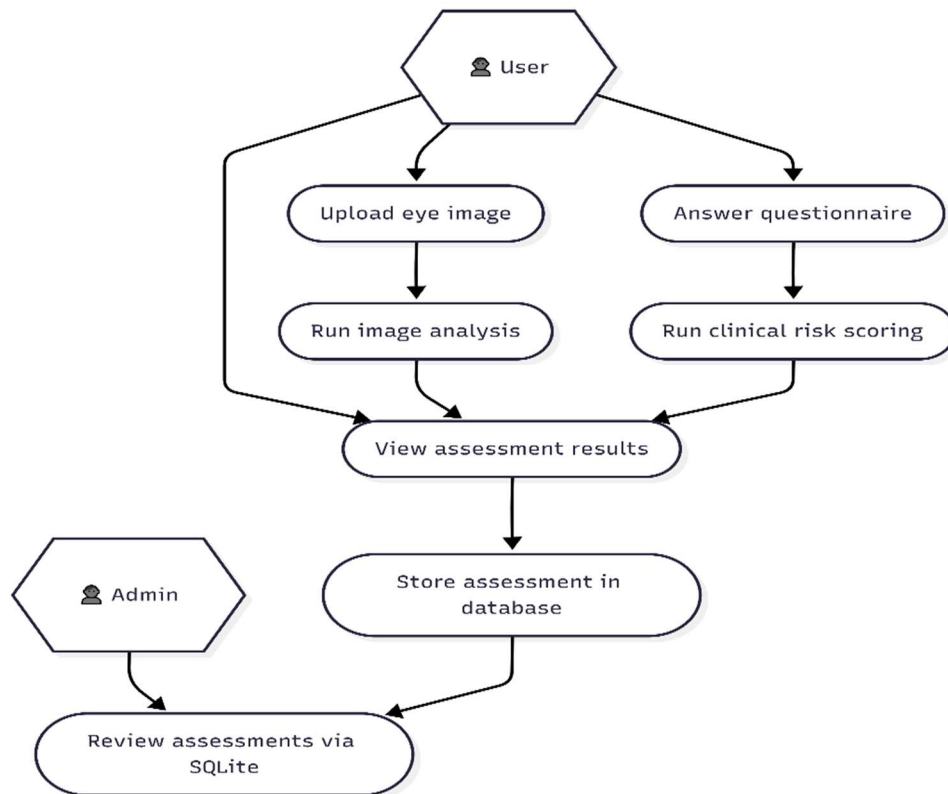


Fig 4 UseCase Diagram

3.3.3 Activity Diagram

The activity diagram illustrates the end-to-end workflow of the proposed ML-based Strabismus Risk Assessment System. The process begins when the user starts the assessment from the home page, uploads or captures an eye image, and enters clinical details through a questionnaire. The system then preprocesses the input data and analyzes the image using a CNN model while simultaneously evaluating the questionnaire data using a machine learning model. The outputs from both models are combined using rule-based decision logic to determine the final diagnosis, risk level, and recommendations. The results are stored in a SQLite database and displayed to the user, with an option to return to the home page for a new assessment.

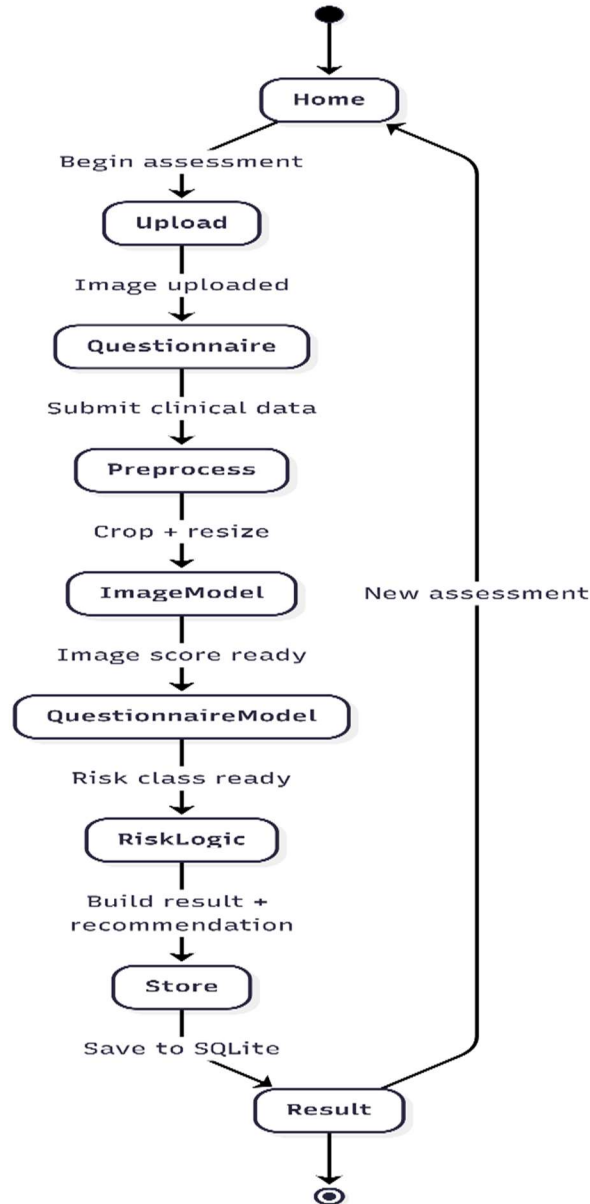


Fig 5 Activity Diagram

3.3.4 Sequence Diagram

Details the step-by-step interaction over time. The user uploads an image, submits clinical data, the system preprocesses the image, runs both models, applies decision rules, saves the assessment, and finally returns the results page

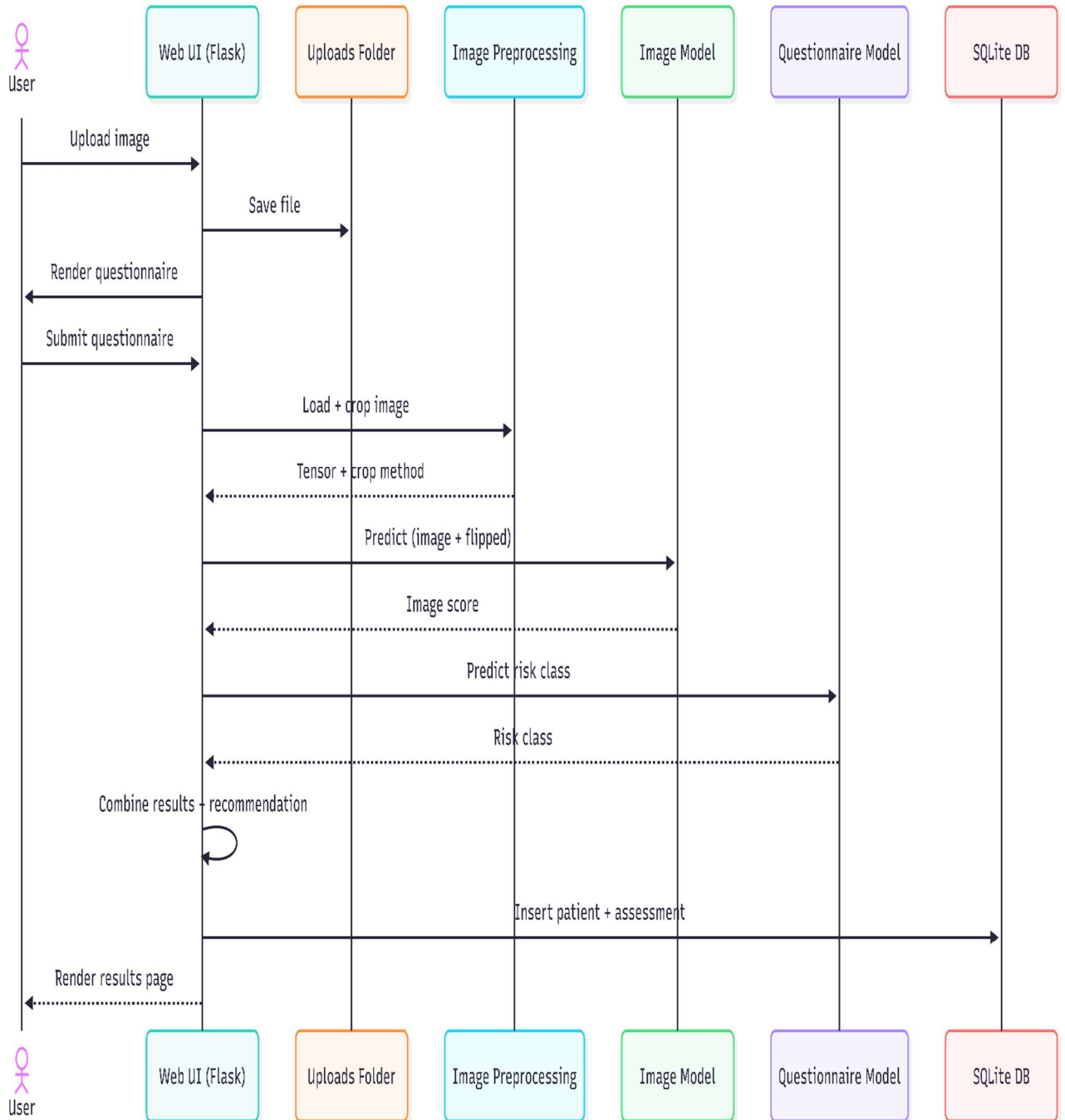


Fig 6 Sequence Diagram

3.3.5 Class Diagram

Describes the system's static structure. It shows the Flask app routes, database helper functions, preprocessing logic, the image and questionnaire model components, and the core data entities (Patient and Assessment) with their relationships.

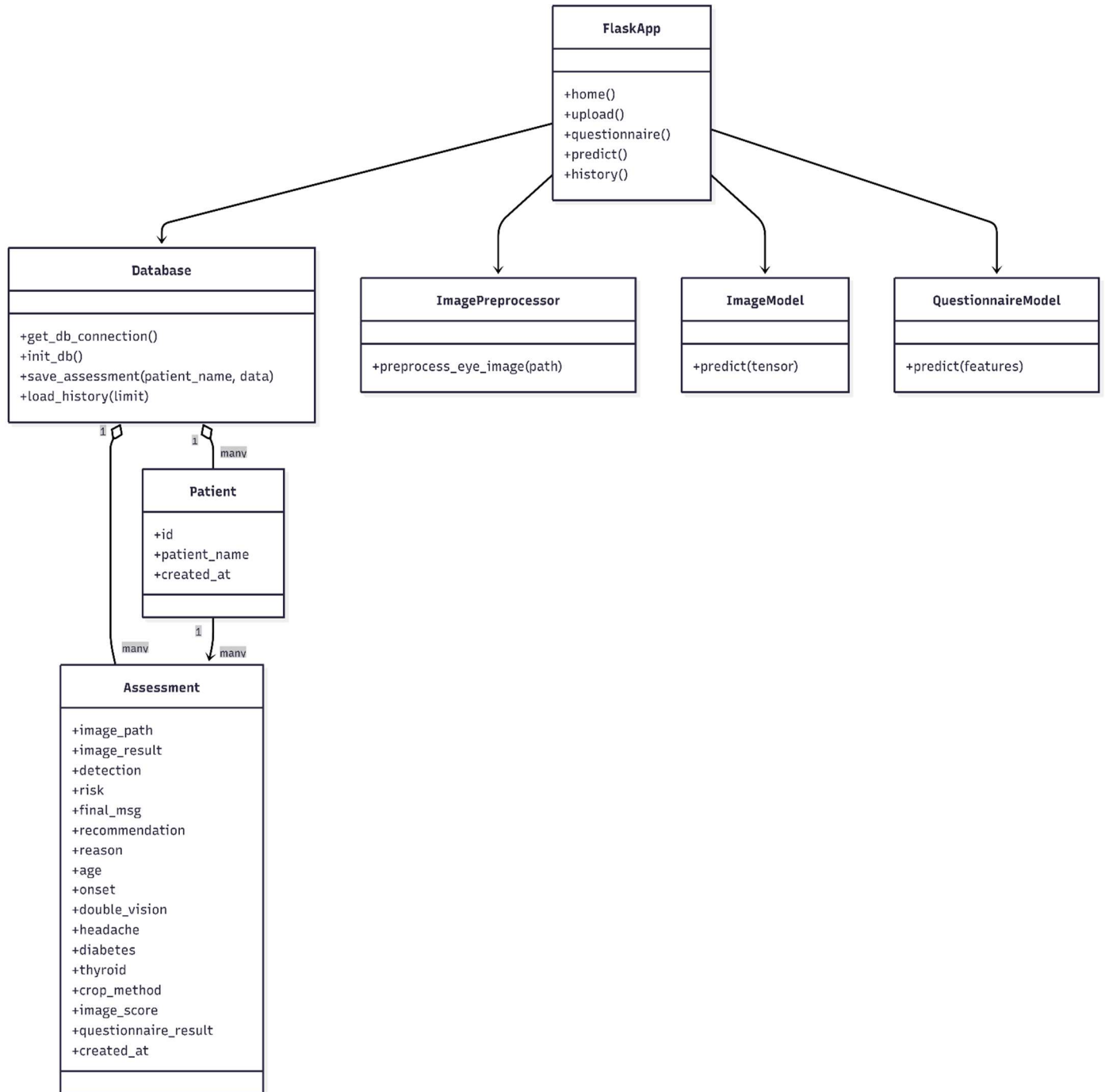


Fig 7 Class Diagram

3.4 METHODOLOGY

Step 1: Data Acquisition

The system begins by collecting input data from the user through a Flask-based web interface. The user provides an eye image either by uploading or capturing it, along with clinical questionnaire details. This step ensures that both visual and medical data are available for further analysis.

Step 2: Input Validation

In this step, the system verifies whether the provided inputs are valid before processing. It checks the image format (JPG/PNG), ensures that no required fields are left empty, and validates clinical responses. This validation is implemented using conditional checks in the Flask backend to avoid errors during prediction.

Step 3: Image Preprocessing

The uploaded eye image is prepared for model prediction by applying preprocessing techniques. The image is resized to 224×224 pixels, pixel values are normalized to a range between 0 and 1, and the image is converted into an array format. Tools like OpenCV or PIL are used for resizing, while NumPy is used for array conversion and normalization.

Step 4: Image-Based Prediction

After preprocessing, the image is passed into a Convolutional Neural Network model for analysis. The system uses MobileNetV2 with transfer learning to extract features and classify the image. The output identifies whether the eye condition is normal or indicates strabismus, along with a probability score.

Step 5: Clinical Data Collection

The system collects additional health-related information from the user through a form interface. This includes parameters such as age, onset type, presence of double vision, headaches, diabetes, and thyroid issues. This data provides important context for improving prediction accuracy.

Step 6: Clinical Data Preprocessing

The collected clinical data is transformed into a format suitable for machine learning models. Categorical values such as Yes/No are converted into binary values (1/0), and numerical values are scaled using normalization techniques like Min-Max scaling. Encoding methods such as label encoding or binary encoding are used in this step.

Step 7: Clinical Risk Prediction

The processed clinical data is analyzed using a Logistic Regression model to estimate the risk of strabismus. This supervised learning model produces a probability score indicating the likelihood of the condition based on medical parameters.

Step 8: Fusion of Model Outputs

The outputs obtained from both the CNN model and Logistic Regression model are combined using conditional rule-based logic: if the image predicts strabismus (both forward and flipped predictions ≥ 0.80) AND the questionnaire indicates risk (not LOW) OR the image shows very high confidence (≥ 0.999), then strabismus is confirmed. This prevents false positives and ensures both visual and clinical data are considered.

Step 9: Risk Classification

Based on the rule-based decision, the system classifies the risk into categories such as low, moderate, or high. This classification is performed using predefined threshold values, making the result easier to interpret for users.

Step 10: Result Generation

The system generates a comprehensive output that includes the detection result, risk level, possible reasons based on symptoms, and recommendations for further action. This step uses rule-based logic to convert model outputs into meaningful insights.

Step 11: Result Display

Finally, the results are presented to the user through a web interface. Using Flask and HTML, the system renders the output in a clear and user-friendly format, ensuring easy understanding of the diagnosis and recommendations.

FLOWCHART :

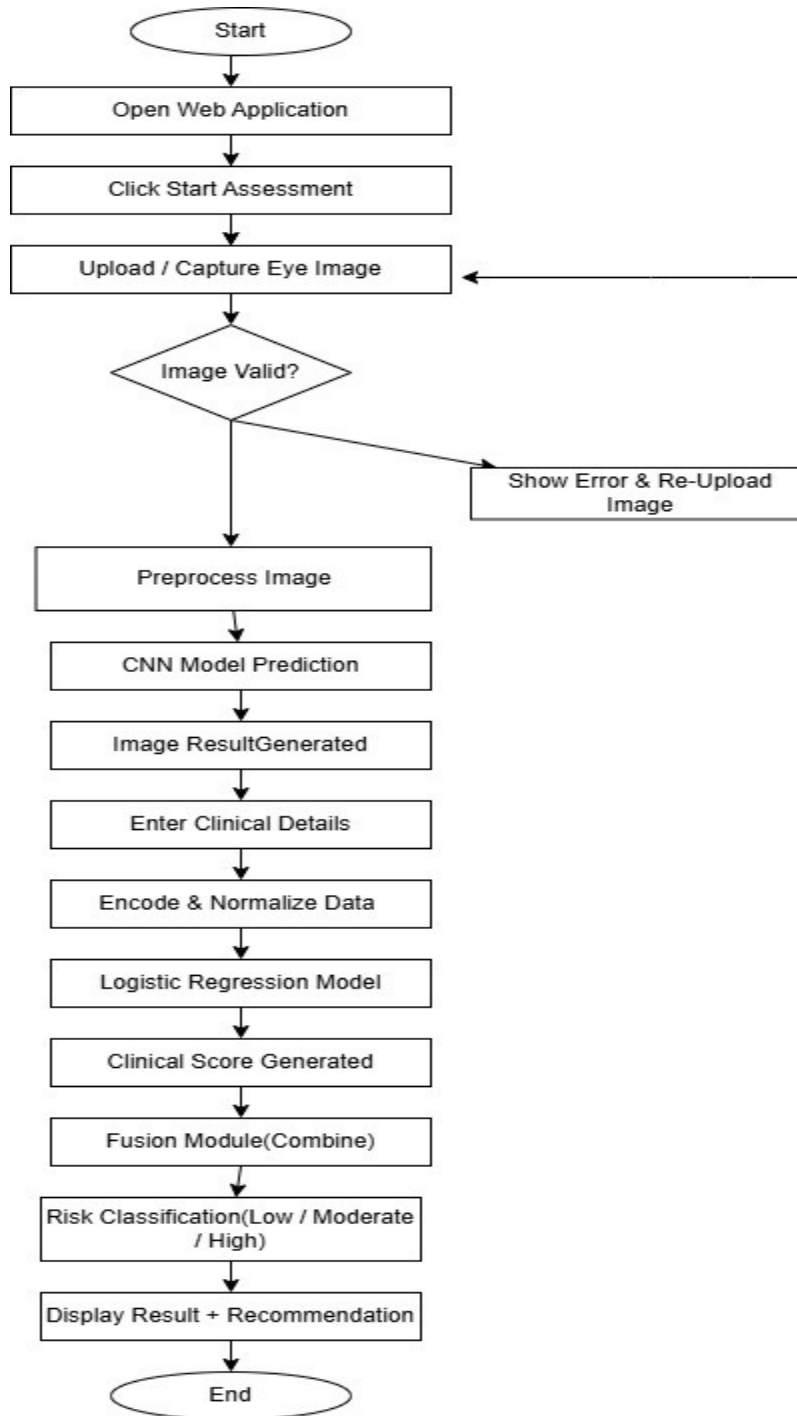


Fig 2 Flowchart

3.5 MODULE DESCRIPTION

Module 1 – Input Data Module

The input data module is responsible for collecting all necessary information required for the system. It gathers the frontal face (eye) image through upload or camera capture and collects clinical details through a structured questionnaire. This module ensures that both visual and clinical inputs are available for further processing, forming the foundation for the system's prediction process.

Module 2 – Web Application Module

The web application module acts as an interface between the user and the backend system. It manages data collection, validates the inputs provided by the user, and ensures that the data is in the correct format. In addition, this module performs normalization and encoding operations to convert raw input into a structured and machine-readable form, enabling smooth data flow into the processing stage.

Module 3 – Data Processing Module

The data processing module prepares both image and clinical data for model prediction. The image preprocessing step includes resizing the input image to a fixed dimension of 224×224 pixels and normalizing pixel values to maintain consistency across all inputs. At the same time, the questionnaire processing step converts categorical responses into numerical values using encoding techniques and scales numerical features such as age. This ensures that the data is compatible with machine learning models.

Module 4 – Model Server Module

The model server module is responsible for performing the core prediction tasks using trained machine learning models, loaded directly within the Flask web application (not as a separate server). The Convolutional Neural Network (CNN), specifically MobileNetV2, analyzes the preprocessed eye image to detect strabismus and generate a probability score. Simultaneously, the Logistic Regression model processes the encoded clinical data to estimate the risk level based on the user's symptoms and medical history. Both models operate independently to produce their respective outputs.

Module 5 – Fusion Module

The fusion module integrates the outputs obtained from the CNN model and the Logistic Regression model. It applies a threshold-based rule: strabismus is detected only if the image prediction is positive ($\text{pred} \geq 0.80$ for both forward and flipped) AND either the questionnaire risk is not LOW or the image confidence exceeds 0.999. This conditional logic prevents false positives when clinical factors are minimal, thereby improving accuracy and reliability of the overall system.

Module 6 – Output Module

The output module is responsible for presenting the final results to the user in a clear and understandable format. It displays the detection result indicating whether strabismus is present or not, provides the risk classification such as low, moderate, or high, and includes explanations and recommendations based on the user's input data. This module ensures that the results are easily interpretable by the user.

The database module stores the generated results and relevant user data for future reference and analysis. It enables the system to maintain records of predictions, which can be used for monitoring, evaluation, and further improvements. This module supports data persistence and enhances the scalability of the system.

CHAPTER 4

IMPLEMENTATION & RESULTS

4.1 TOOLS & TECHNOLOGIES USED

The strabismus risk assessment system runs on a standard x64 workstation. For development and deployment, a dual-core CPU, 4–8 GB RAM, and 10–20 GB free storage are sufficient because inference uses lightweight models and runs on CPU. A GPU is optional and only needed to speed up training, not for routine assessment. Stable internet access is required only for updates or data transfer, not for local inference.

Table 2 : Hardware Requirements

Component	Minimum Requirement	Recommended
Processor	Dual-core x64, 2.0 GHz	Quad-core 3.0+ GHz
RAM	4 GB	8–16 GB
Storage	10 GB free	20+ GB SSD
Network	Standard broadband	Stable broadband (uploading images)
GPU (optional)	Not required	Entry-level CUDA GPU for faster training

The software stack is fully Python-based. The Flask framework hosts the web app, TensorFlow performs image inference, OpenCV handles eye-region preprocessing, and scikit-learn performs questionnaire risk prediction. SQLite is used for local persistence of patient assessments, while joblib loads the saved model and scaler. HTML/CSS templates provide the user interface.

Table 3 : Software Requirement

Software	Version	Purpose
Python	3.10+	Core runtime
Flask	3.x	Web application server
TensorFlow	2.x	Image model inference
OpenCV	4.x	Image preprocessing and eye detection
scikit-learn	1.x	Questionnaire model
joblib	Latest	Model and scaler loading
SQLite	встроено	Local assessment history storage
HTML/CSS/JS	Any	UI templates

4.2 SYSTEM IMPLEMENTATION

The implementation followed an iterative, module-by-module approach aligned with the system's dependency chain. The core runtime pipeline was implemented first in the Flask application, because it defines the end-to-end behavior and exposes the primary user workflow. This includes image intake, eye-region preprocessing, image inference, questionnaire inference, and risk fusion, all hosted in `app.py`. Once the runtime pipeline was stable, the model training and evaluation scripts were finalized to produce the deployable artifacts used at inference time, including the MobileNetV2 image classifier and the questionnaire logistic regression model (see `train_mobilenetv2.py`, `train_questionnaire_model.py`, and `evaluate_model.py`). The dataset pipeline was implemented last, providing reproducible data preparation, augmentation, and splitting utilities for both image and questionnaire data (see `data_augmentation.py`, `img_dataset_split.py`, `Qn_dataset_creation.py`, and `Qn_dataset_split.py`).

Three implementation patterns are worth highlighting. First, the image preprocessing stage uses a tiered fallback strategy: it attempts face-plus-eye detection, then global eye detection, and finally a centered upper-face crop to ensure a usable region of interest even under poor capture conditions. This improves robustness and avoids brittle failures in the image pipeline (see `app.py`). Second, risk estimation is modeled as a fusion of two independent predictors: the image classifier and the questionnaire classifier. The decision logic applies a guard condition that mitigates false positives, requiring high image confidence when the questionnaire indicates low risk. This prevents over-triggering when clinical signals do not support the visual prediction (see `app.py`). Third, persistence is built around a simple, auditable SQLite log. Each assessment is stored with image path, model outputs, questionnaire responses, and derived recommendations, enabling historical review at the operator level (see `app.py`).

```
const upload = new FormData();
upload.append("image", imageFile);
await fetch(`${baseUrl}/questionnaire`, { method: "POST", body: upload });
const form = new FormData();
form.append("img_path", "uploads/your_uploaded_image.jpg");
form.append("age", "22"); form.append("onset", "1"); form.append("double_vision", "0");
form.append("headache", "0"); form.append("diabetes", "0"); form.append("thyroid", "0");
const res = await fetch(`${baseUrl}/predict`, { method: "POST", body: form });
const html = await res.text();
```

4.3 WORKING MODEL (STEP-BY-STEP EXECUTION)

Step 1: System Initialization

The user accesses the web application through the home screen, which introduces the system and its purpose. By clicking the "Begin Assessment" button, the user initiates the strabismus detection process and is guided to the next stage of input collection.

Step 2: Eye Image Acquisition

The system prompts the user to provide an eye image either by uploading from the device or capturing it using a camera. A preview of the selected image is displayed, and validation checks are performed to ensure that a proper image is provided before proceeding further.

Step 3: Clinical Data Collection

After confirming the image, the system collects clinical details through a structured questionnaire. The user inputs information such as age, onset type, and symptoms including double vision, headache, diabetes, and thyroid issues, which are essential for clinical risk evaluation.

Step 4: Input Validation and Preprocessing

The system verifies the completeness and correctness of both image and clinical inputs. The image is resized to a standard dimension (224×224) and normalized, while the questionnaire data is converted into numerical format using encoding techniques and scaled appropriately for model processing.

Step 5: Image-Based Prediction using CNN

The preprocessed eye image is passed to the trained Convolutional Neural Network (MobileNetV2), which analyzes eye alignment and extracts visual features. Based on this analysis, the model generates a probability score indicating the presence or absence of strabismus.

Step 6: Clinical Risk Prediction using Logistic Regression

Simultaneously, the encoded clinical data is provided to the Logistic Regression model. This model evaluates the risk level based on symptoms and medical conditions, producing a clinical probability score that reflects the likelihood of strabismus.

Step 7: Fusion of Model Outputs

The outputs obtained from both the CNN model and Logistic Regression model are combined using conditional rule-based logic: if the image predicts strabismus (both forward and flipped predictions ≥ 0.80) AND the questionnaire indicates risk (not LOW) OR the image shows very high confidence (≥ 0.999), then strabismus is confirmed. This prevents false positives and ensures both visual and clinical data are considered.

Step 8: Risk Categorization and Decision Making

Based on the rule-based decision, the system determines whether strabismus is detected or not. The risk level (low, moderate, or high) is independently generated from the questionnaire model (not dependent on image detection). The final detection result and risk level are presented together, allowing the risk classification to inform the recommendation even if strabismus is not detected.

Step 9: Result Display and Recommendation

The final output is presented to the user in a clear format, including detection results, risk level, possible reasons derived from input data, and recommendations such as medical consultation or routine monitoring.

Step 10: Data Storage for Future Reference

The system stores the prediction results and relevant input data in a local database, enabling future analysis, tracking, and system improvement.

4.4 RESULTS & OUTPUTS

Home Page

Fig 8 shows the home page of the ML-Based Strabismus Risk Assessment System. It provides an overview of the system and allows users to initiate the assessment process by clicking the "Begin Assessment" button.

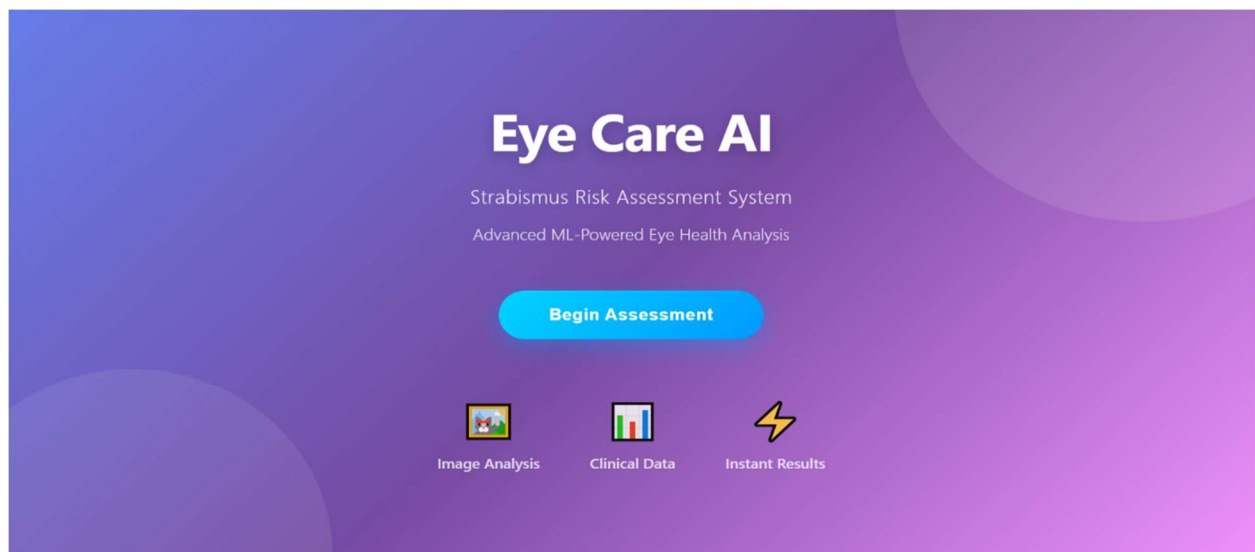
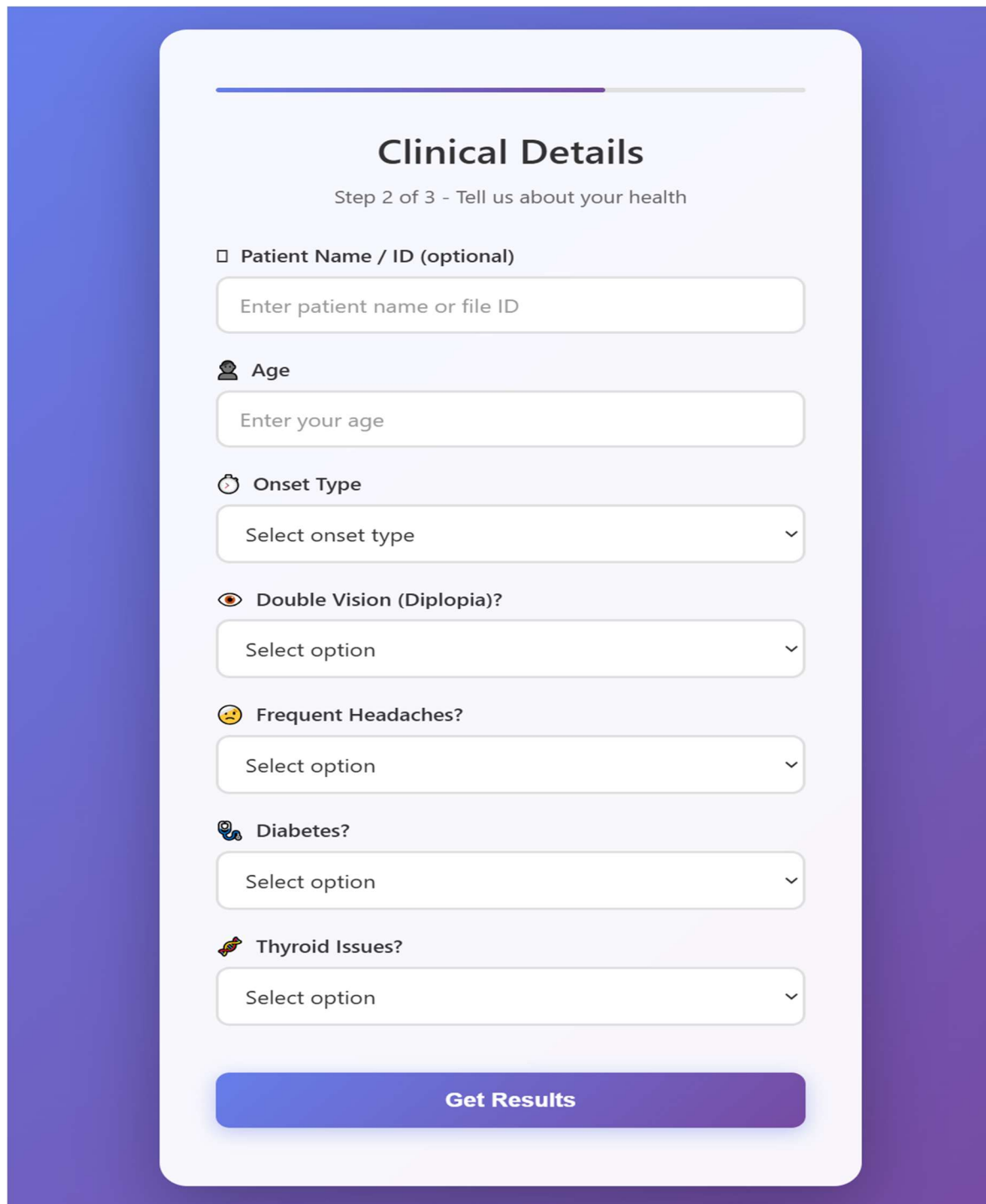


Fig 8 Home Page

Clinical Details Page

Fig 9 shows the clinical details input page where users provide additional medical information such as age, onset type, double vision, headache, diabetes, and thyroid issues. This data is used for risk prediction.



The image shows a mobile application interface for a clinical details input page. The page has a light purple background with a darker purple gradient at the bottom. The title "Clinical Details" is centered at the top in a bold, black font. Below the title, a subtitle "Step 2 of 3 - Tell us about your health" is displayed. The form consists of several input fields, each with a corresponding icon and label. The first field is for "Patient Name / ID (optional)" with a document icon. The second field is for "Age" with a person icon. The third field is for "Onset Type" with a clock icon. The fourth field is for "Double Vision (Diplopia)?" with an eye icon. The fifth field is for "Frequent Headaches?" with a sad face icon. The sixth field is for "Diabetes?" with a stethoscope icon. The seventh field is for "Thyroid Issues?" with a thyroid gland icon. Each field has a placeholder text "Select option" and a dropdown arrow. At the bottom of the form is a large, rounded rectangular button with a gradient from light purple to dark purple, labeled "Get Results".

Clinical Details

Step 2 of 3 - Tell us about your health

☐ Patient Name / ID (optional)

Enter patient name or file ID

 Age

Enter your age

 Onset Type

Select onset type

 Double Vision (Diplopia)?

Select option

 Frequent Headaches?

Select option

 Diabetes?

Select option

 Thyroid Issues?

Select option

Get Results

Fig 9 Clinical details page

Image Upload Page

Fig 10 shows the image upload interface, where users can either upload an existing eye photo or capture a new one. The selected image is previewed for verification, allowing users to confirm clarity, alignment, and correctness before proceeding. The page also provides basic guidance on acceptable image quality and framing, helping reduce errors and improve diagnostic accuracy. This step ensures the input is valid and user-approved before analysis

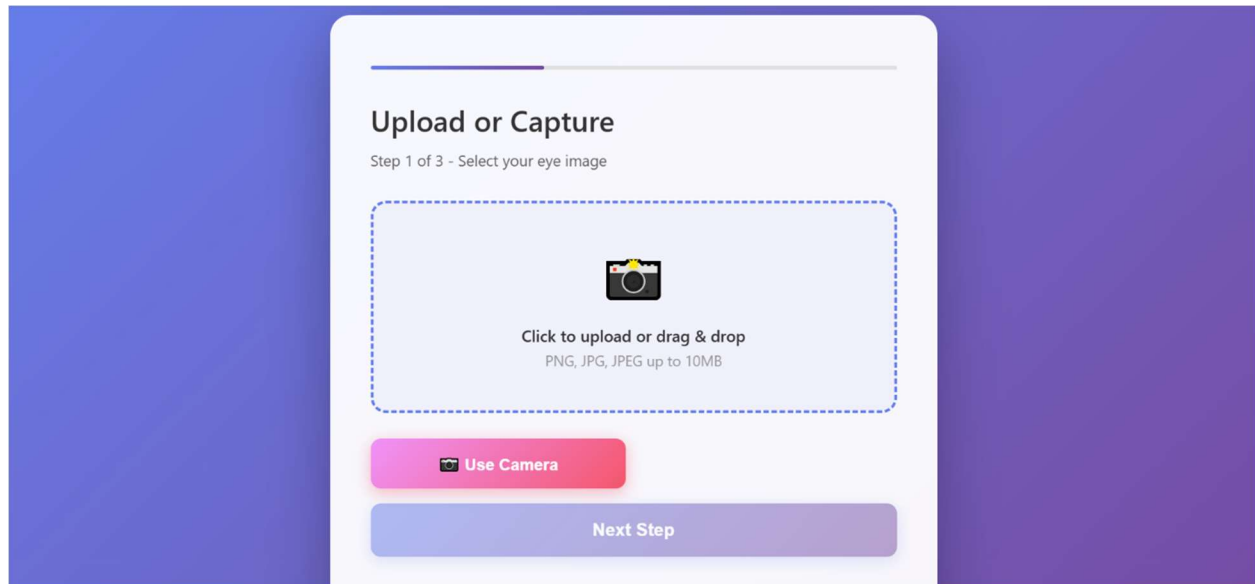


Fig !0 Image Upload Page

Result Page

Fig 11 shows the result page displaying the analysis outcome. The system identifies whether strabismus is present, calculates the risk level, explains the decision with clinical input factors, and suggests recommended next steps. It also summarizes key questionnaire responses, shows confidence or severity cues when available, and guides the user on whether to seek specialist evaluation or continue routine monitoring. The page is designed to be clear, actionable, and easy to interpret for both patients and clinicians.

Assessment Complete

Step 3 of 3 - Your Results
Patient: Patient-b1dd89fa

Your Eye Image



Image Analysis

Classification:
Strabismus Eye

Strabismus Status: **Detected**

Clinical Risk Assessment

Overall Risk Level:

MODERATE

Contributing Factors

- Double vision present, Frequent headaches

Medical Recommendation

Eye check-up recommended within a week.

STRABISMUS PRESENT

[← New Assessment](#)

⚠ Medical Disclaimer: This AI-assisted system is designed for screening purposes only and should not be used as a substitute for professional medical diagnosis. Please consult with a qualified ophthalmologist for comprehensive eye evaluation and treatment.

Fig 11 Result Page

4.4.1 GRAPHS

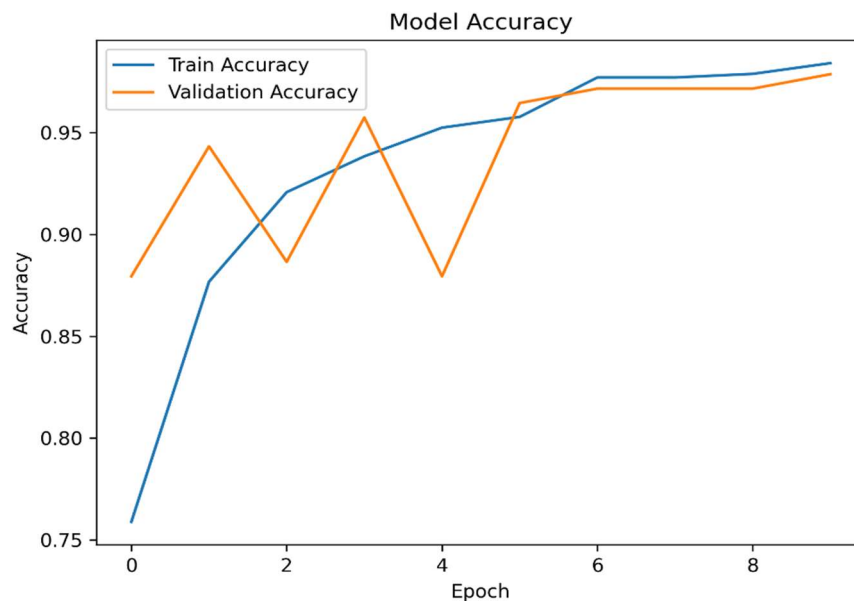


Fig 12 IIMAGE ACCURACY CURVE

This plot shows how training accuracy and validation accuracy change across epochs for the image model. The two curves indicate how well the model is learning from the training data and how that learning generalizes to unseen validation images. A small gap between the curves suggests good generalization, while a widening gap can indicate overfitting.

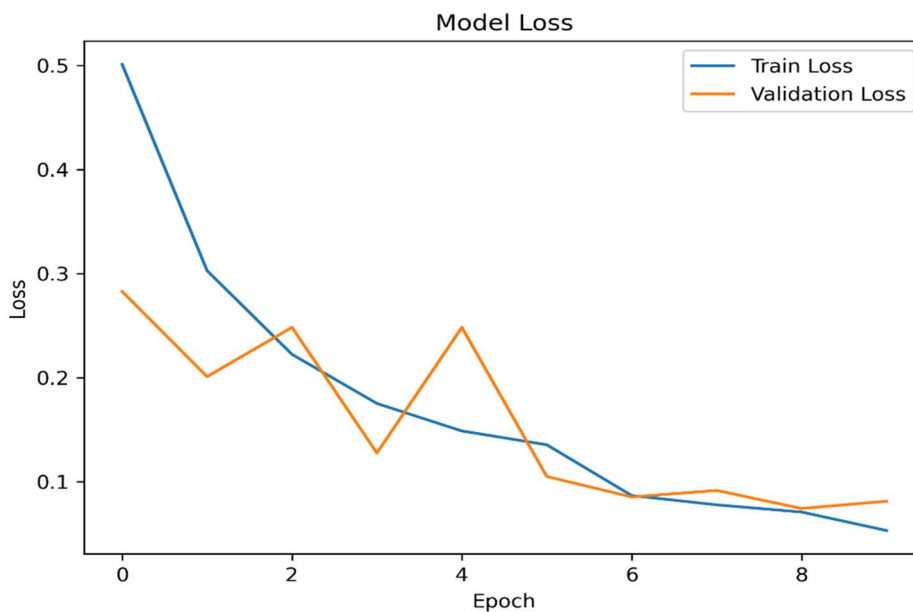


Fig 13 IIMAGE LOSS CURVE

This plot shows training loss and validation loss over epochs for the image model. Loss reflects how far predictions are from the true labels, so a downward trend indicates improvement. If validation loss stops decreasing or rises while training loss continues to fall, it signals overfitting.

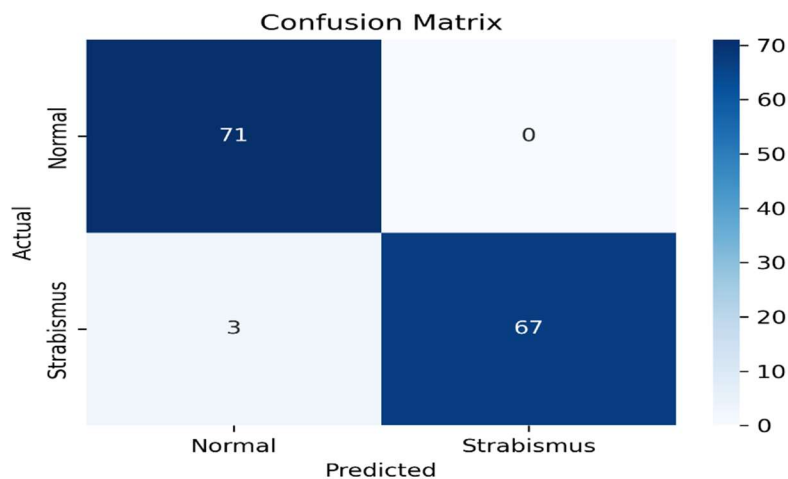


Fig 14 IMAGE CONFUSION MATRIX

The confusion matrix summarizes image model predictions by class (Normal vs Strabismus). The diagonal cells show correct classifications, while off-diagonal cells show misclassifications. This plot helps identify which class is more often confused and provides a quick view of overall classification quality.

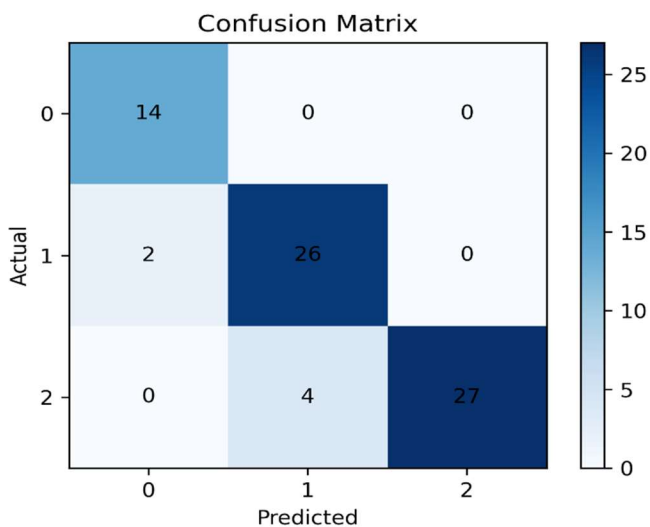


Fig 15 QUESTIONNAIRE CONFUSION MATRIX

This confusion matrix shows how the questionnaire model classifies each risk category. Correct predictions are on the diagonal; errors appear off the diagonal. It highlights which risk levels are most often misclassified and whether the model tends to over- or under-predict certain classes.

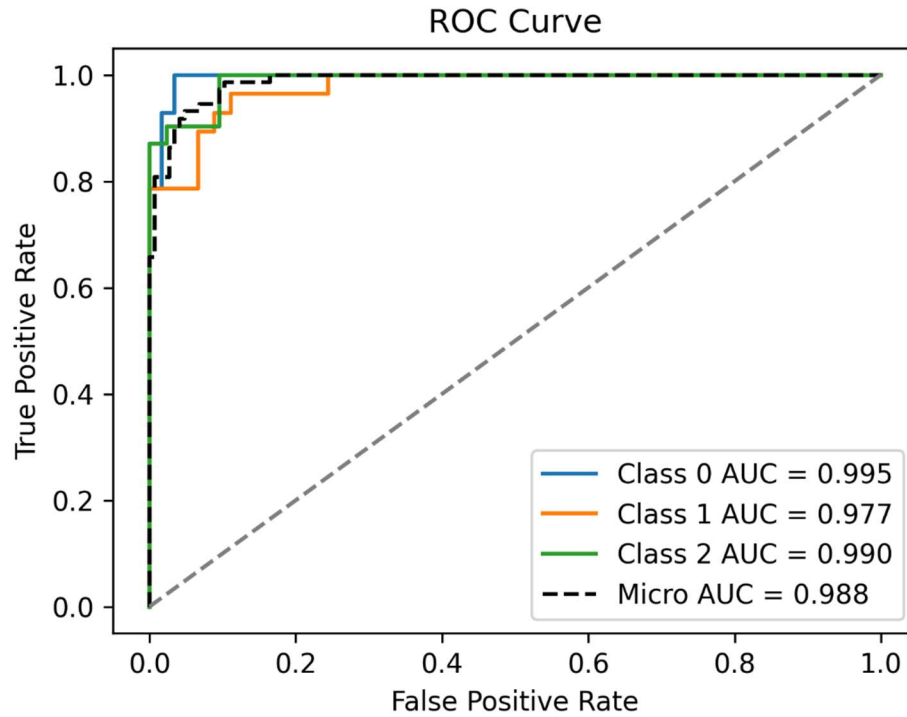


Fig 16 QUESTIONAIRE ROC CURVE

This ROC curve shows the trade-off between true positive rate and false positive rate for the questionnaire model (one-vs-rest for each class, with a micro-average). Curves closer to the top-left indicate better discrimination. The AUC values quantify how well the model separates each class from the others.

4.5 PERFORMANCE ANALYSIS

1. Performance Metrics Table (Image Model)

The overall performance of the image-based model is summarized using key evaluation metrics:

Metric	Value
Test Accuracy	97.87%
Macro Precision	0.98
Macro Recall	0.98
Macro F1-Score	0.98
Weighted F1-Score	0.98

Table 4: Performance Metrics Table (Image Model)

These values indicate that the model achieves high accuracy and maintains a strong balance between precision and recall. The high F1-score confirms reliable performance for both classes (Normal vs Strabismus).

2. Class-wise Performance Analysis (Image Model)

The system performs consistently across both classes:

Class	Precision	Recall	F1-Score	Support
Normal	0.96	1.00	0.98	71
Strabismus	1.00	0.96	0.98	70
Macro Avg	0.98	0.98	0.98	141
Weighted Avg	0.98	0.98	0.98	141

Table 5: Class-wise Performance Analysis (Image Model)

The results show strong and balanced classification across both categories, with minimal confusion between Normal and Strabismus.

3. Performance Metrics Table (Questionnaire Model)

The overall performance of the questionnaire model is summarized using key evaluation metrics:

Metric	Value
Test Accuracy	91.78%
Macro Precision	0.91
Macro Recall	0.93
Macro F1-Score	0.92
Weighted F1-Score	0.92

Table 6: Performance Metrics Table (Questionnaire Model)

These values show good predictive quality across multiple risk classes, with balanced precision and recall.

4. Class-wise Performance Analysis (Questionnaire Model)

The system performs consistently across all questionnaire risk classes:

Class	Precision	Recall	F1-Score	Support
0	0.88	1.00	0.93	14
1	0.87	0.93	0.90	28
2	1.00	0.87	0.93	31
Macro Avg	0.91	0.93	0.92	73
Weighted Avg	0.92	0.92	0.92	73

Table 7: Class-wise Performance Analysis (Questionnaire Model)

Even the lowest-performing class maintains strong precision and recall, indicating robust classification across risk levels.

5. Overall Interpretation

The graphical and tabular analysis confirms that the Strabismus Risk Assessment system achieves high performance with:

- High accuracy for the image-based classifier
- Strong classification across all questionnaire risk levels
- Balanced precision and recall, indicating low bias toward any class
- Consistent F1-scores, showing reliable decision boundaries

The combined use of image analysis and questionnaire-based prediction improves overall assessment reliability and supports early risk identification.

6. Comparison with Existing Practice

Compared to conventional screening approaches that rely primarily on visual inspection or limited patient information, the proposed system integrates image-based features with structured questionnaire data. This provides more consistent, data-driven predictions and improves reliability across different cases. The machine learning-based approach captures complex relationships in the data and produces measurable performance metrics, which are typically unavailable in manual assessment methods.

CHAPTER 5

5.1 CONCLUSION

The Strabismus Risk Assessment System provides an effective and intelligent solution for early detection of strabismus using both eye image analysis and clinical questionnaire data. By integrating Deep Learning (MobileNetV2) and Machine Learning techniques, the system delivers fast and reliable predictions.

The system reduces dependency on manual clinical screening by providing results within seconds, improving accessibility and saving time. It minimizes human errors and ensures consistent evaluation through automated processing. The user-friendly interface allows easy interaction for users, making the system suitable for real-time applications.

Overall, the system acts as a cost-effective, scalable, and efficient preliminary screening tool that supports early diagnosis and helps in timely medical intervention.

5.2 FUTURE ENHANCEMENT

The proposed system can be further enhanced by incorporating advanced features to improve its performance and usability.

- **Strabismus Type Classification:**

The system can be extended to classify different types of strabismus such as Esotropia, Exotropia, Hypertropia, and Hypotropia, instead of only detecting the presence of the condition. This will provide more detailed diagnostic information.

- **Mobile Application Development:**

A mobile application can be developed to enable real-time eye analysis using a smartphone camera. This will improve accessibility and allow users to perform screening easily from anywhere.

- **Improved Accuracy with Advanced Models:**

The system can be enhanced by using advanced deep learning models such as EfficientNet or Vision Transformers along with larger datasets. This will improve prediction accuracy, robustness, and real-world performance.

APPENDICES

APPENDIX A: MAIN APPLICATION (app.py)

```
import os, cv2, numpy as np
from uuid import uuid4
from flask import Flask, render_template, request
import tensorflow as tf, joblib

app = Flask(__name__)
UPLOAD_FOLDER = "static/uploads"
os.makedirs(UPLOAD_FOLDER, exist_ok=True)
app.config["UPLOAD_FOLDER"] = UPLOAD_FOLDER

# Load Models
image_model = tf.keras.models.load_model("strabismus_mobilenetv2_model.h5")
questionnaire_model = joblib.load("questionnaire_model.pkl")
scaler = joblib.load("questionnaire_scaler.pkl")

THRESHOLD = 0.80

🔍 Image Preprocessing (Reduced)
def preprocess_eye_image(path):
    img = cv2.imread(path)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    face = cv2.CascadeClassifier(cv2.data.harcascades + "haarcascade_frontalface_default.xml")
    eyes = cv2.CascadeClassifier(cv2.data.harcascades + "haarcascade_eye.xml")

    faces = face.detectMultiScale(gray, 1.1, 5)

    if len(faces) > 0:
        x, y, w, h = faces[0]
        roi = img[y:y+h, x:x+w]
    else:
        roi = img

    roi = cv2.resize(roi, (224, 224)) / 255.0
    return np.expand_dims(roi, axis=0)

🌐 Routes
@app.route("/")
def home():
    return render_template("home.html")

@app.route("/upload")
```

```
def upload():
    return render_template("upload.html")
```

⬆️ Upload Handling

```
@app.route("/questionnaire", methods=["POST"])
def questionnaire():
    file = request.files["image"]
    name = f"{uuid4().hex}.jpg"
    path = os.path.join(app.config["UPLOAD_FOLDER"], name)
    file.save(path)
    return render_template("questionnaire.html", img_path=f"uploads/{name}")
```

□ Prediction Logic

```
@app.route("/predict", methods=["POST"])
def predict():
    img_path = request.form["img_path"]
    full = os.path.join("static", img_path)

    img = preprocess_eye_image(full)

    p1 = image_model.predict(img)[0][0]
    p2 = image_model.predict(np.flip(img, axis=2))[0][0]
    pred = (p1 + p2) / 2

    detected = p1 >= THRESHOLD and p2 >= THRESHOLD

    image_result = "Strabismus Eye" if detected else "Normal Eye"
    detection = "Detected" if detected else "Not Detected"
```

📋 Questionnaire Model

```
age = int(request.form["age"])
onset = int(request.form["onset"])
dv = int(request.form["double_vision"])
hd = int(request.form["headache"])
db = int(request.form["diabetes"])
th = int(request.form["thyroid"])

features = scaler.transform([[age, onset, dv, hd, db, th]])
risk = ["LOW", "MODERATE", "HIGH"][questionnaire_model.predict(features)[0]]
```

□ Output Logic

```
final_msg = "STRABISMUS PRESENT" if detected else "NO STRABISMUS"
```

```

reasons = []
if dv: reasons.append("Double vision")
if hd: reasons.append("Headache")
if db: reasons.append("Diabetes")
if th: reasons.append("Thyroid")

reason = ", ".join(reasons) if reasons else "No major risk factors"

recommendation = {
    "HIGH": "Consult immediately",
    "MODERATE": "Check within a week",
    "LOW": "Routine monitoring"
}[risk]

return render_template("result.html",
    img_path=img_path,
    image_result=image_result,
    detection=detection,
    risk=risk,
    reason=reason,
    recommendation=recommendation,
    final_msg=final_msg
)

```

APPENDIX B: HTML (SIMPLIFIED)

🏠 B1: Home Page

```

<h1>Eye Care AI</h1>
<p>Strabismus Detection System</p>

<form action="/upload">
    <button>Start</button>
</form>

```

📁 B2: Upload Page

```

<form action="/questionnaire" method="post" enctype="multipart/form-data">
    <input type="file" name="image" required>
    <button>Next</button>
</form>

```


B3: Questionnaire Page

```
<form action="/predict" method="post">
<input type="hidden" name="img_path" value="{{ img_path }}">

<input type="number" name="age" required>

<select name="onset">
<option value="0">Birth</option>
<option value="1">Gradual</option>
<option value="2">Sudden</option>
</select>

<select name="double_vision">
<option value="0">No</option>
<option value="1">Yes</option>
</select>

<select name="headache">
<option value="0">No</option>
<option value="1">Yes</option>
</select>

<select name="diabetes">
<option value="0">No</option>
<option value="1">Yes</option>
</select>

<select name="thyroid">
<option value="0">No</option>
<option value="1">Yes</option>
</select>

<button>Submit</button>
</form>
```

B4: Result Page

```
<h2>Result</h2>



<p><b>Image Result:</b> {{ image_result }}</p>
<p><b>Status:</b> {{ detection }}</p>
<p><b>Risk:</b> {{ risk }}</p>

<p><b>Factors:</b> {{ reason }}</p>
```

<p>Recommendation: {{ recommendation }}</p>

<h3>{{ final_msg }}</h3>

<form action="/">

<button>New Test</button>

</form>

APPENDIX C: TRAINING FILES

Include as-is:

- train_mobilenetv2.py
- train_questionnaire_model.py

APPENDIX D: DATA PIPELINE

Include:

- image preprocessing
- augmentation
- dataset split
- questionnaire dataset

APPENDIX E: PROJECT STRUCTURE

Project/

```
|
|— app.py
|— templates/
|— models/
|— preprocessing/
|— training/
|— static/uploads
```

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